CSE201: Advanced Programming

AP Test Summary: University ERP

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1. Test Plan

1.1 Overview

This Test Plan outlines the functional, negative, UI, and security tests carried out for the University ERP System. The purpose is to validate that all major workflows (Admin, Instructor, and Student) function correctly. All tests were performed using a local MySQL instance populated with a test dataset.

1.2 Test Environment

- Operating System: Windows 11 / macOS
- JDK Version: OpenJDK 21
- Database: MySQL 8.0
- Dependencies: Java Swing, FlatLaf, MigLayout, MySQL Connector/J, HikariCP, jBCrypt, OpenCSV.

2. Acceptance Tests

2.1 Login & Authentication

Test Case	Expected Outcome	Status
Login Admin	Admin Dashboard opens	PASS
Login Student	Student Dashboard opens	PASS
Login Instructor	Instructor Dashboard opens	PASS
Wrong Password	Error: "Incorrect password"	PASS
Empty Fields	Error: "Username empty"	PASS
5 Failed Attempts	Account locked (2 min)	PASS

2.2 Student Module

Feature	Action	Status
Course Catalog	Catalog loads active sections	PASS
Registration	Adds to enrollment DB	PASS
Register (Full)	Blocked if capacity is 0	PASS
Drop Section	Status changes to DROPPED	PASS
Transcript	CSV file generated	PASS

2.3 Instructor Module

Feature	Action	Status
View Sections	Only assigned sections visible	PASS
Enter Grades	Scores saved to DB	PASS
Compute Final	Weighted avg calculated	PASS
Statistics	Avg/Max/Min displayed	PASS

2.4 Admin Module

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Feature	Action	Status
User Mgmt	Create Student/Instructor	PASS
Rollback	DB cleanup on failure	PASS
Course Mgmt	Create Course/Section	PASS
Conflict Check	Room/Time conflicts blocked	PASS

3. Test Summary

Category	Status	Remarks
Login & Auth	PASS	Role-based routing functional
Student Flows	PASS	Catalog, Drop, CSV Export working
Instructor Flows	PASS	Grading & Schema logic verified
Admin Flows	PASS	Two-phase creation working
Maintenance Mode	PASS	Global write-block functional
Data Integrity	PASS	Unique constraints enforced
Code Org	PASS	Strict UI/Service separation

Conclusion:

The system meets all functional requirements and passed all critical acceptance tests defined in the Project Brief.